



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No.: 4

1. Title: Use function types, scope, and closures; apply try-catch; build a palindrome checker.

2. Software/Tools Required: Visual Studio Code (VS Code) Google Chrome/web browser, HTML, JavaScript.

3. Theory:

Function Types: Demonstrates anonymous functions (assigned directly to variables without a name) and arrow functions (concise ES6 => syntax).

Scope: Differentiates between **global variables** (accessible throughout the script) and **local variables** (restricted to the block or function where they are declared).

Closures: Occurs when an inner function retains access to its parent function's scope, preserving state variables (like `count`) even after the outer function finishes executing.

Error Handling: Uses `try...catch` and `throw` statements to validate input types and gracefully handle runtime exceptions without crashing the program.



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Palindrome Algorithm: A string is split into an array of characters, reversed, joined back into a string, and compared against the original input using strict equality (===).

4. Output:

```
1 //Palindrome Checker
2 //Name: Sonal Nakhate PRN: 24070521199
3 //Scope, closure, error handling using try-catch
4
5 //Create anonymous function without a name
6
7 let welcome = function() {
8     console.log("Palindrome Checker Program.");
9 }
10
11 welcome();
12
13 //Global Variable
14 let language = "JavaScript";
15
16 function ScopeDemo() {
17     //Local Variable
18     let topic = "Functions";
19     console.log("Language:", language); //Display global var
```

Topic: Functions
Counter: 1
Counter: 2
madam is a Palindrome.
PS C:\Users\Sonal\Desktop\Documents\SEM V\JS_24070521199> node Exp4\Palindrome.js
Palindrome Checker Program.
Language: JavaScript
Topic: Functions
Counter: 1
Counter: 2
madam is a Palindrome.
PS C:\Users\Sonal\Desktop\Documents\SEM V\JS_24070521199>

5. Result/Conclusion: In this experiment, fundamental JavaScript concepts—including anonymous/arrow functions, scope rules, closures, and `try...catch` error handling—were successfully implemented. Using these techniques alongside array manipulation methods (`split`, `reverse`, `join`), a functional and error-resistant palindrome checker was constructed.



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Case Study No.: 4

1. Case Study Title: ATM Card Pin Verification using JavaScript Functions Case Study.

2. Case Study Program Code:

```
// Global Scope: System configurations
//Name: Sonal Nakhate PRN: 24070521199
const BANK_NAME = "Global Secure Bank";
let dailyAttemptsLimit = 3;

// 1. FUNCTION DECLARATION: Reverses a PIN string or number
function reversePin(pin) {
  const pinString = String(pin);
  const reversed = pinString.split("").reverse().join("");
  return reversed;
}

// 2. FUNCTION EXPRESSION: Checks if a PIN is a palindrome
const isPalindromePin = function(pin) {
  const pinString = String(pin);
  const reversedPin = reversePin(pinString);
  return pinString === reversedPin;
};

// 3. ARROW FUNCTION: Formats output messages based on verification
const generateSecurityAlert = (pin, isPalindrome) => {
  if (isPalindrome) {
    return `[SPECIAL SECURITY PROTOCOL]: PIN ${pin} is a Palindrome.
High-symmetry alert triggered. Proceeding to main menu...`;
  }
}
```



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```
    return `[STANDARD SECURITY]: PIN ${pin} verified successfully.
    Proceeding to main menu...`;
};

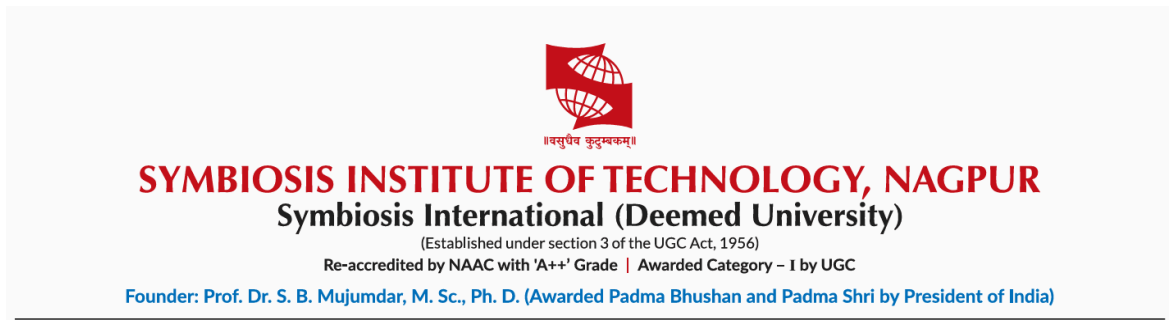
// 4. SCOPE DEMONSTRATION: Verification Handler
function processATMTransaction(enteredPin) {
    let attemptCounter = 1;

    console.log(`--- Processing PIN verification at ${BANK_NAME} ---`);

    if (attemptCounter <= dailyAttemptsLimit) {
        const isPalindromic = isPalindromePin(enteredPin);
        const securityMessage = generateSecurityAlert(enteredPin,
isPalindromic);
        console.log(securityMessage);

        // FIXED OBJECT RETURN: Comma added after true
        return {
            success: true,
            isPalindrome: isPalindromic
        };
    } else {
        console.log("Account locked. Exceeded daily limits.");
        return {
            success: false,
            isPalindrome: false
        };
    }
}

// Execution
processATMTransaction(1221);
processATMTransaction(4821);
```



3. Output:

```
1 // Global Scope: System configurations
2 //Name: Sonal Nakhate PRN: 24070521199
3 const BANK_NAME = "Global Secure Bank";
4 let dailyAttemptsLimit = 3;
5
6 // 1. FUNCTION DECLARATION: Reverses a PIN string or number
7 function reversePin(pin) {
8   const pinString = String(pin);
9   const reversed = pinString.split("").reverse().join("");
10  return reversed;
11 }
12
13 // 2. FUNCTION EXPRESSION: Checks if a PIN is a palindrome
14 const isPalindromePin = function(pin) {
15   const pinString = String(pin);
16   const reversedPin = reversePin(pinString);
17   return pinString === reversedPin;
18 };
19
```

C:\Program Files\nodejs\node.exe .\CS-ATMpin.js
--- Processing PIN verification at Global Secure Bank ---
[SPECIAL SECURITY PROTOCOL]: PIN 1221 is a Palindrome. High-symmetry alert triggered. Proceeding to main menu...
--- Processing PIN verification at Global Secure Bank ---
[STANDARD SECURITY]: PIN 4821 verified successfully. Proceeding to main menu...

4. Result/Conclusion: We have successfully implemented a simple JavaScript program that checks if the ATM pin is a palindrom or not.